

Executive Summary

Predicting Customer Credit Card Default Risk from Historical Financial Data(Classification)

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Why This Matters

Credit card issuers extend revolving credit and carry the risk that a cardholder will fail to make a required payment. If that risk can be anticipated before the payment is due, a financial institution can intervene early — for example, by adjusting a credit limit, sending targeted reminders, or beginning proactive outreach — rather than reacting only after a client has already defaulted.

This project asks a specific, decision-oriented question: using a client's demographics, credit limit, and the last six months of billing and repayment history, can we identify which clients are likely to default on next month's payment? The intended user is a credit risk or collections team that needs to prioritize accounts for review, not a fully automated approval or denial system.

Data and Target

The analysis uses the UCI “Default of Credit Card Clients” dataset (I-Cheng Yeh, 2016): 30,000 Taiwanese credit card accounts from 2005, with 25 attributes per client. Each row is one account holder, and the target is a binary flag — default payment next month (1) vs. no default (0).

Predictors used: credit limit, age, sex, education, marital status, the six most recent months of repayment status, bill amounts, and payment amounts, plus a small set of engineered features (credit-utilization ratio, payment-coverage ratio, worst recent delinquency, and total delinquent months).

Class balance: about 22% of clients defaulted (6,636 of 30,000) and 78% did not — a meaningful imbalance that makes plain accuracy a misleading scorecard (a model that always guesses “no default” is already 78% “accurate” while catching zero actual defaulters).

Key limitations of the data: the dataset is a single historical snapshot of Taiwan in 2005, two demographic fields contained a small number of undocumented codes that were cleaned before modeling, and no information from after the target month was permitted into the model (verified through an explicit leakage audit).

Methods Compared

All data was split into training (70%), validation (15%), and test (15%) sets, preserving the 22% default rate in each. Every preprocessing step (scaling, category encoding) was fit only on the training data to prevent information from leaking into model evaluation. Four models were compared:

- Majority-class baseline — always predicts “no default”; establishes the floor every real model must beat.
- Logistic Regression — an interpretable linear model, useful for showing which factors move the prediction and in which direction.
- Random Forest — a tree-based ensemble capable of capturing non-linear patterns and interactions among variables.
- Gradient Boosting — a second, typically stronger tree-based ensemble that builds trees sequentially to correct earlier errors.

Each model (other than the baseline) was tuned with cross-validation, and its decision threshold was separately optimized on the validation set to give each model its best fair chance rather than judging all of them at a generic 50% cutoff. The deciding metric throughout was F1-score on the Default class — a single number that balances catching real defaulters (recall) against not over-flagging good clients (precision), since both types of error carry real business cost.

Main Results

The table below compares each tuned model on the validation set, at its own optimal decision threshold — the basis on which the final model was selected. The winning model was then evaluated exactly once on a separate, untouched test set to confirm the choice held up (see Selected Model, below).

Model	Accuracy	Precision	Recall	F1	ROC AUC
Baseline (majority class)	77.9%	0.00	0.00	0.000	—
Logistic Regression (tuned)	80.3%	0.556	0.548	0.552	0.771
Random Forest (tuned)	78.7%	0.516	0.594	0.552	0.785
Gradient Boosting (tuned) — SELECTED	81.2%	0.580	0.537	0.557	0.788

All three real models landed close together (F1 scores within about 0.006 of each other), comfortably ahead of the baseline's F1 of 0.00. Gradient Boosting had the best combination of F1 and ROC AUC (a measure of how well the model ranks defaulters above non-defaulters across all possible thresholds; 0.788 is well above the 0.5 “coin flip” level) and was carried forward as the final model.

Most important diagnostic finding: permutation importance on the held-out test set confirms that recent repayment delinquency — specifically, how many months behind a client currently is — is by far the strongest predictor, a signal independently confirmed by the exploratory analysis, the Logistic Regression coefficients, and the Random Forest importances. Demographic fields (sex, education, marital status, age) contributed comparatively little on their own.

Selected Model

The final model is Gradient Boosting, tuned and evaluated at its own optimal decision threshold. On the held-out test set it reaches an F1 of 0.544 and ROC AUC of 0.778 for identifying next-month defaulters — a substantial improvement over the baseline's F1 of 0.00. It was selected because it offered the best overall trade-off between correctly flagging defaulters and avoiding false alarms, plus the strongest ranking ability of the three tuned candidates, though its lead over Random Forest and Logistic Regression was narrow rather than decisive.

At this threshold, the model correctly identifies about 53% of true defaulters (recall) and, of the accounts it flags as high-risk, about 56% actually go on to default (precision). In practical terms: it is a substantial early-warning signal, not a certainty, and should be used to prioritize attention rather than to make final decisions on its own.

Recommendations

- Use the model's output as a ranking/prioritization signal for manual review or proactive outreach on high-risk accounts — not as an automated approve/deny decision.
- Direct review capacity toward clients showing recent repayment delinquency first, since this is the dominant risk driver identified across every stage of the analysis.
- Maintain a complementary review process (e.g., periodic sampling) for clients with no delinquency history, since the model is largely blind to defaults that occur without a prior delinquency trail (see Limitations).
- Conduct a fair-lending / disparate-impact review before any operational deployment, since sex, education, and marital status were included as model inputs even though their individual contribution to the final prediction was modest.
- Re-validate the model on more recent data before relying on it operationally, since the training data reflects a single market and year (Taiwan, 2005).

Limitations

- Data: single historical snapshot (Taiwan, 2005); repayment behavior and credit norms may differ in other markets or time periods, and performance is not guaranteed to hold without re-validation.
- Modeling: the selected model still misses close to half of actual defaulters and roughly 44% of its default flags are false alarms — a meaningful improvement over the baseline, but far from perfect.
- Validation: the three tuned candidate models were statistically close in performance (F1 within ~ 0.006), so a different validation sample could plausibly favor a different model; the choice of Gradient Boosting should be treated as a reasonable, not uniquely correct, selection.
- Interpretation: every relationship in this analysis is a predictive association learned from historical data, not a proven cause-and-effect relationship — the model should not be used to assert that any factor causes default.
- Generalization: missed defaulters (false negatives) look statistically similar to correctly classified non-defaulters on the features the model relies on most, meaning the model offers little protection against defaults with no prior warning signs in the data it was given.